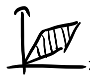
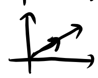
线性代数

行列式的定义与性质

定义一（本质）：$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12} = S_{\square OABC}$ 平行四边形面积

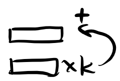
$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ = 平行六面体体积

$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} \neq 0$ ，线性无关

$\begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$  线性相关

$\begin{vmatrix} 1 & 2 \\ 0 & 0 \end{vmatrix} = 0$ 

性质

① 转置，

⇒ det 不变

② 零行，两行成比例 ⇒ det=0

③ 倍乘 $\begin{vmatrix} \ddots & & & \\ ka & kb & kc & kd \\ & & & \ddots \end{vmatrix} = k \begin{vmatrix} \ddots & & & \\ a & b & c & d \\ & & & \ddots \end{vmatrix}$

eg.  比如面积翻倍，只用翻倍一条边

④ 单行可拆 $\begin{vmatrix} \ddots & & & \\ a_1+b_1 & a_2+b_2 & a_3+b_3 & a_4+b_4 \\ & & & \ddots \end{vmatrix} = \begin{vmatrix} \ddots & & & \\ a_1 & a_2 & a_3 & a_4 \\ & & & \ddots \end{vmatrix} + \begin{vmatrix} \ddots & & & \\ b_1 & b_2 & b_3 & b_4 \\ & & & \ddots \end{vmatrix}$

⑤ 两行(列)互换 ⇒ det 变号

定义、逆序数法

排列　n个数构成一个n级排列

逆序　大在前小在后叫一个逆序

eg $\tau(621534) = 8$　62 61 65 63 64 / 21 53 54

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \sum_{j_1 j_2 \cdots j_n} (-1)^{\tau(j_1 j_2 \cdots j_n)} \cdot a_{1j_1} a_{2j_2} \cdots a_{nj_n}$$

eg $|a_{11}| = a_{11}$

eg $\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{22}$

eg $\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{14} & a_{15} \\ a_{21} & a_{22} & a_{23} & a_{24} & a_{25} \\ a_{31} & a_{32} & a_{33} & a_{34} & a_{35} \\ a_{41} & a_{42} & a_{43} & a_{44} & a_{45} \\ a_{51} & a_{52} & a_{53} & a_{54} & a_{55} \end{vmatrix}$　确定 $a_{12}a_{31}a_{54}a_{43}a_{25}$ 前的符号

$= a_{12}a_{25}a_{31}a_{43}a_{54} \cdot \underbrace{(-1)^{\tau(25134)}}_{(-1)^4 = 1}$

n阶行列式有 n! 项

eg $\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$　$a_{1a}a_{2b}a_{3c}$

abc
- 1 2 3 → τ=0
- 1 3 2 → τ=1
- 2 1 3 → τ=1
- 2 3 1 → τ=2
- 3 1 2 → τ=2
- 3 2 1 → τ=3

定义 行列式的展开定理 → 降阶

1. 余子式

   去掉 a_{ij} 所在行与列剩下的是 a_{ij} 的余子式,叫 M_{ij}

2. 代数余子式

   $A_{ij}=(-1)^{i+j}\cdot M_{ij}$; $M_{ij}=(-1)^{i+j}A_{ij}$

3. 行列式按行(列)展开

$$|A|=\begin{cases} a_{i1}A_{i1}+a_{i2}A_{i2}+\cdots+a_{in}A_{in} \\ a_{1i}A_{1i}+a_{2i}A_{2i}+\cdots+a_{ni}A_{ni} \end{cases}$$

例 1-1

例 1.1 $D_4=\begin{vmatrix} 2 & -1 & 0 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & 2 & -1 \\ -1 & 0 & 0 & 2 \end{vmatrix}=(\quad)$.

$$D_4=2\cdot(-1)^{1+1}\cdot\begin{vmatrix} 2 & -1 & 0 \\ 0 & 2 & -1 \\ 0 & 0 & 2 \end{vmatrix}+(-1)(-1)^{4+1}\begin{vmatrix} -1 & 0 & 0 \\ 2 & -1 & 0 \\ 0 & 2 & -1 \end{vmatrix}$$

$$=2^4+(-1)^3=15$$

几个重要的行列式

1. 主对角行列式: 对角线元素相乘

$$\begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ 0 & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix}=\begin{vmatrix} a_{11} & 0 & \cdots & 0 \\ a_{21} & a_{22} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}=\begin{vmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix}=\prod_{i=1}^{n}a_{ii}.$$

2. 副对角行列式: $(-1)^{\frac{n(n-1)}{2}}\cdot$ 对角线元素相乘

$$\begin{vmatrix} a_{11} & \cdots & a_{1,n-1} & a_{1n} \\ a_{21} & \cdots & a_{2,n-1} & 0 \\ \vdots & & \vdots & \vdots \\ a_{n1} & \cdots & 0 & 0 \end{vmatrix}=\begin{vmatrix} 0 & \cdots & 0 & a_{1n} \\ 0 & \cdots & a_{2,n-1} & a_{2n} \\ \vdots & & \vdots & \vdots \\ a_{n1} & \cdots & a_{n,n-1} & a_{nn} \end{vmatrix}=\begin{vmatrix} 0 & \cdots & 0 & a_{1n} \\ 0 & \cdots & a_{2,n-1} & 0 \\ \vdots & & \vdots & \vdots \\ a_{n1} & \cdots & 0 & 0 \end{vmatrix}$$

$$=(-1)^{\frac{n(n-1)}{2}}a_{1n}a_{2,n-1}\cdots a_{n1}.$$

$\tau(n,n-1,\cdots,2,1)=(n-1)+(n-2)+\cdots+1$

$=\frac{n(n-1)}{2}$

3. 拉普拉斯展开式 (A 是 m 阶, B 是 n 阶)

$$\begin{vmatrix} A & O \\ O & B \end{vmatrix}=\begin{vmatrix} A & C \\ O & B \end{vmatrix}=\begin{vmatrix} A & O \\ C & B \end{vmatrix}=|A||B|,$$

$$\begin{vmatrix} O & A \\ B & O \end{vmatrix}=\begin{vmatrix} C & A \\ B & O \end{vmatrix}=\begin{vmatrix} O & A \\ B & C \end{vmatrix}=(-1)^{mn}|A||B|.$$

4. 范德蒙德行列式

"高年级欺负低年级,欺负遍了"

盯着第二行

$$\begin{vmatrix} 1 & 1 & \cdots & 1 \\ \boxed{x_1 \quad x_2 \quad \cdots \quad x_n} \\ x_1^2 & x_2^2 & \cdots & x_n^2 \\ \vdots & \vdots & & \vdots \\ x_1^{n-1} & x_2^{n-1} & \cdots & x_n^{n-1} \end{vmatrix}=\prod_{1\leqslant i<j\leqslant n}(x_j-x_i).$$

行列式的计算

1
1. 化基本形法　直接展开　（某行/列有最多的 0）　eg 例1.1
2. 爪型　用斜边消去另一边　eg 例1.2
3. 异爪型　阶数不高：直接展开，阶数高：递推法　习1.5　eg 例1.3，习1.5
4. 行(列)和相等：所有列(行)加，提公因式　eg 例1.4
5. 消零化基本形
6. 拉普拉斯展开式　$\begin{vmatrix} A & O \\ O & B \end{vmatrix} = |A||B|$，$\begin{vmatrix} O & A \\ B & O \end{vmatrix} = (-1)^{mn}|A||B|$　eg 例1.5
7. 范德蒙行列式：盯着第二行　eg 例1.6

2. 递推法：建立 D_n 与 D_{n-1} 的关系式，实现递推　eg 例1.7
- 元素分布规律相同
- D_{n-1} 只比 D_n 少一阶

例1.3

行列式 $\begin{vmatrix} \lambda & -1 & 0 & 0 \\ 0 & \lambda & -1 & 0 \\ 0 & 0 & \lambda & -1 \\ 4 & 3 & 2 & \lambda+1 \end{vmatrix} = 4(-1)^{1+4}\begin{vmatrix} -1 & 0 & 0 \\ \lambda & -1 & 0 \\ 0 & \lambda & -1 \end{vmatrix} + \lambda(-1)^{1+1}\begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 3 & 2 & \lambda+1 \end{vmatrix}$

异爪型

$$= 4 + \lambda\left[3(-1)^{1+3}\begin{vmatrix} -1 & 0 \\ \lambda & -1 \end{vmatrix} + \lambda(-1)^{1+1}\begin{vmatrix} \lambda & -1 \\ 2 & \lambda+1 \end{vmatrix}\right]$$

$$= 4 + \lambda\left[3 + \lambda(\lambda^2+\lambda+2)\right]$$

$$= \lambda^4 + \lambda^3 + 2\lambda^2 + 3\lambda + 4$$ （繁）

按最下一行展开 $= 4(-1)^{1+4}\begin{vmatrix} -1 & 0 & 0 \\ \lambda & -1 & 0 \\ 0 & \lambda & -1 \end{vmatrix} + 3(-1)^{2+4}\begin{vmatrix} \lambda & 0 & 0 \\ 0 & -1 & 0 \\ 0 & \lambda & -1 \end{vmatrix} + 2(-1)^{3+4}\begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & -1 \end{vmatrix} + (\lambda+1)(-1)^{4+4}\begin{vmatrix} \lambda & -1 & 0 \\ 0 & \lambda & -1 \\ 0 & 0 & \lambda \end{vmatrix}$

$$= 4 + 3\lambda + 2\lambda^2 + (\lambda+1)\lambda^3$$

例1.2

计算行列式：$\begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 0 \\ 1 & 0 & 3 & 0 \\ 1 & 0 & 0 & 4 \end{vmatrix}$. $= 2\times3\times4\begin{vmatrix} 1 & 1 & 1 & 1 \\ \frac{1}{2} & 1 & & \\ \frac{1}{3} & & 1 & \\ \frac{1}{4} & & & 1 \end{vmatrix} = 2\times3\times4\begin{vmatrix} 1-\frac{1}{2}-\frac{1}{3}-\frac{1}{4} & 0 & 0 & 0 \\ \frac{1}{2} & 1 & & \\ \frac{1}{3} & & 1 & \\ \frac{1}{4} & & & 1 \end{vmatrix} = 2\times3\times4\times(1-\frac{1}{2}-\frac{1}{3}-\frac{1}{4})\times1\times1\times1$

爪型

例1.4 ☆

$$D_n = \begin{vmatrix} a & b & b & \cdots & b \\ b & a & b & \cdots & b \\ b & b & a & \cdots & b \\ \vdots & \vdots & \vdots & & \vdots \\ b & b & b & \cdots & a \end{vmatrix}.$$

行和相等列相加

$$D_n = \begin{vmatrix} a+(n-1)b & b & b & \cdots & b \\ a+(n-1)b & a & b & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a+(n-1)b & b & b & \cdots & a \end{vmatrix} = (a+(n-1)b)\begin{vmatrix} 1 & b & b & \cdots & b \\ 1 & a & b & \cdots & b \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & b & b & \cdots & a \end{vmatrix} = (a+(n-1)b)\begin{vmatrix} 1 & b & b & \cdots & b \\ 1 & a-b & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & 0 & 0 & \cdots & a-b \end{vmatrix} = [a+(n-1)b]\cdot(a-b)^{n-1}$$

【注3】这类字母抽象型行列式显然具有代表性.如

(1)当 $a=0,b=1$ 时，$\begin{vmatrix} 0 & 1 & 1 & \cdots & 1 \\ 1 & 0 & 1 & \cdots & 1 \\ 1 & 1 & 0 & \cdots & 1 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 1 & 1 & \cdots & 0 \end{vmatrix}_{n\times n}=(n-1)(-1)^{n-1}$；

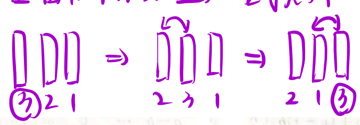
当 $a=2,b=1$ 时，$\begin{vmatrix} 2 & 1 & 1 & \cdots & 1 \\ 1 & 2 & 1 & \cdots & 1 \\ 1 & 1 & 2 & \cdots & 1 \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & 1 & 1 & \cdots & 2 \end{vmatrix}_{n\times n}=n+1.$

(2)当 $a=x$ 时，$\begin{vmatrix} x & b & b & \cdots & b \\ b & x & b & \cdots & b \\ b & b & x & \cdots & b \\ \vdots & \vdots & \vdots & & \vdots \\ b & b & b & \cdots & x \end{vmatrix}_{n\times n}=[x+(n-1)b](x-b)^{n-1}.$

若视 x 为变量，b 是常数，则行列式是 x 的 n 次多项式，其根是 $x_1=x_2=\cdots=x_{n-1}=b,x_n=(1-n)b.$

当 a 取为 $\lambda-a$ 时，$\begin{vmatrix} \lambda-a & b & b & \cdots & b \\ b & \lambda-a & b & \cdots & b \\ b & b & \lambda-a & \cdots & b \\ \vdots & \vdots & \vdots & & \vdots \\ b & b & b & \cdots & \lambda-a \end{vmatrix}_{n\times n}=[\lambda-a+(n-1)b](\lambda-a-b)^{n-1}.$

如果 a在副对角线上，要换列：



(3)当 a 在副对角线上时，$G_n=\begin{vmatrix} b & b & \cdots & b & a \\ b & b & \cdots & a & b \\ \vdots & \vdots & & \vdots & \vdots \\ b & a & \cdots & b & b \\ a & b & \cdots & b & b \end{vmatrix}\overset{(*)}{=\!=}(-1)^{\frac{n(n-1)}{2}}\begin{vmatrix} a & b & \cdots & b & b \\ b & a & \cdots & b & b \\ \vdots & \vdots & & \vdots & \vdots \\ b & b & \cdots & a & b \\ b & b & \cdots & b & a \end{vmatrix}$

$$=(-1)^{\frac{n(n-1)}{2}}[a+(n-1)b](a-b)^{n-1}.$$

(*)处是将最后1列和前面相邻列对换，对换 $n-1$ 次到第1列，再将最新的行列式的最后1列和相邻列对换，对换 $n-2$ 次到第2列，……，直到换成 D_n，共交换 $(n-1)+(n-2)+\cdots+1=\frac{n(n-1)}{2}$ 次，故得 ⟸ 记住 $(-1)^{\frac{n(n-1)}{2}}$

$$G_n=(-1)^{\frac{n(n-1)}{2}}[a+(n-1)b](a-b)^{n-1}.$$

注意，上述结果不要当作公式去记忆，而是要学会分析行列式中元素分布的规律性，并掌握相应的计算方法.

例1.5

$D_4=\begin{vmatrix} a_1 & 0 & 0 & b_1 \\ 0 & a_2 & b_2 & 0 \\ 0 & b_3 & a_3 & 0 \\ b_4 & 0 & 0 & a_4 \end{vmatrix}.$

$$=(-1)\begin{vmatrix} a_1 & b_1 & 0 & 0 \\ 0 & 0 & b_2 & a_2 \\ 0 & 0 & a_3 & b_3 \\ b_4 & a_4 & 0 & 0 \end{vmatrix}=(-1)^2\begin{vmatrix} 0 & 0 & a_3 & b_3 \\ 0 & 0 & b_2 & a_2 \\ a_1 & b_1 & 0 & 0 \\ b_4 & a_4 & 0 & 0 \end{vmatrix}=(-1)^{2\times2}\begin{vmatrix} a_3 & b_3 \\ b_2 & a_2 \end{vmatrix}\begin{vmatrix} a_1 & b_1 \\ b_4 & a_4 \end{vmatrix}$$

例1.6

计算行列式 $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ b+c & a+c & a+b \end{vmatrix}$

$$= \begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ a+b+c & a+b+c & a+b+c \end{vmatrix} = (a+b+c)(-1)^2 \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a+b+c)\left[(b-a)(c-a)(c-b)\right]$$

例1.7

$$D_4 = \begin{vmatrix} 1-a & a & 0 & 0 \\ -1 & 1-a & a & 0 \\ 0 & -1 & 1-a & a \\ 0 & 0 & -1 & 1-a \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & 0 & 0 \\ 0 & 1-a & a & 0 \\ 0 & -1 & 1-a & a \\ -a & 0 & -1 & 1-a \end{vmatrix} = (-a)(-1)^{1+4} a^3 + \begin{vmatrix} 1-a & a & 0 \\ -1 & 1-a & a \\ 0 & -1 & 1-a \end{vmatrix}$$

$$= a^4(-1)^4 + D_3$$

$D_4 = a^4(-1)^4 + D_3$

$D_3 = a^3(-1)^3 + D_2$

$D_2 = a^2(-1)^2 + D_1$

$D_1 = 1-a$

习1.4

$$D_n = \begin{vmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \cdots & 0 & 0 \\ 0 & -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & 0 & \cdots & -1 & 2 \end{vmatrix}_{n\times n}.$$

$$= \begin{vmatrix} 1 & -1 & & & & \\ & 2 & -1 & & 0 & \\ & -1 & 2 & -1 & & \\ & & -1 & 2 & -1 & \\ & & & -1 & 2 & \ddots \\ & & & & \ddots & -1 \\ 1 & & & & -1 & 2 \end{vmatrix}$$

$$= 1\cdot(-1)^{1+n}\cdot 1\cdot(-1)^{n-1} + D_{n-1}$$

$$= (-1)^{2n} + D_{n-1}$$

$$= 1 + D_{n-1}$$

$$= n$$

习1.5

计算 n 阶行列式 $D_n = \begin{vmatrix} a_1 & -1 & 0 & \cdots & 0 & 0 \\ a_2 & x & -1 & \cdots & 0 & 0 \\ a_3 & 0 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a_{n-1} & 0 & 0 & \cdots & x & -1 \\ a_n & 0 & 0 & \cdots & 0 & x \end{vmatrix}$.

$$= a_n\cdot(-1)^{n+1}(-1)^{n-1} + x(-1)^{2n} D_{n-1}$$

$$= a_n + xD_{n-1} = a_n + x(a_{n-1} + xD_{n-2})$$

$D_1 = a_1$

$$\Rightarrow D_n = a_n + xa_{n-1} + x^2 a_{n-2} + \cdots + x^{n-1} a_1$$

行列式表示的函数和方程

例1.8

例 1.8 设 $f(x)=\begin{vmatrix}1&0&x\\1&2&x^2\\1&3&x^3\end{vmatrix}$,求 $f(x+1)-f(x)$.

$$原=\begin{vmatrix}1&0&x+1\\1&2&(x+1)^2\\1&3&(x+1)^3\end{vmatrix}-\begin{vmatrix}1&0&x\\1&2&x^2\\1&3&x^3\end{vmatrix}$$

$$=\begin{vmatrix}1&0&x+1-x\\1&2&x^2+2x+1-x^2\\1&3&x^3+3x^2+3x+1-x^3\end{vmatrix}$$

$$=\begin{vmatrix}1&0&1\\1&2&2x+1\\1&3&3x^2+3x+1\end{vmatrix}=\begin{vmatrix}1&0&1\\0&2&2x\\0&3&3x^2+3x\end{vmatrix}=\begin{vmatrix}1&0&1\\0&2&2x\\0&0&3x^2\end{vmatrix}=6x^2$$

例1.9

例 1.9 设 $f(x)=\begin{vmatrix}x-2&x-1&x-2&x-3\\2x-2&2x-1&2x-2&2x-3\\3x-3&3x-2&4x-5&3x-5\\4x&4x-3&5x-7&4x-3\end{vmatrix}$,则方程 $f(x)=0$ 的根的个数为( ).

$$f(x)=\begin{vmatrix}0&2&x&-3\\0&2&2x&-3\\-x+2&3&4x-2&-x-3\\-x+7&0&5x-7&-x+4\end{vmatrix}=\begin{vmatrix}0&2&x&-1\\0&2&2x&-1\\-x+2&3&3x&-x\\-x+7&0&4x&-x+4\end{vmatrix}$$

$$=-x\begin{vmatrix}0&0&1&0\\0&2&2&1\\-x+2&3&3&x\\-x+7&0&4&x-4\end{vmatrix}=x\begin{vmatrix}0&2&1\\-x+2&3&x\\-x+7&0&x-4\end{vmatrix}=x\begin{vmatrix}0&0&1\\-x+2&-2x+3&x\\-x+7&-2x+8&x-4\end{vmatrix}$$

$$=x\begin{vmatrix}-x+2&-2x+3\\-x+7&-2x+8\end{vmatrix}=5x\begin{vmatrix}-x+2&-2x+3\\1&1\end{vmatrix}=5x(x-1)$$

2个

例1.10

例 1.10 设关于 λ 的方程

$$\begin{vmatrix}\lambda-1&-2&3\\1&\lambda-4&3\\-1&a&\lambda-5\end{vmatrix}=0$$

有二重根,求参数 a 的值.

$$\begin{vmatrix}\lambda-1&-2&3\\1&\lambda-4&3\\-1&a&\lambda-5\end{vmatrix}=\begin{vmatrix}\lambda-2&-\lambda+2&0\\1&\lambda-4&3\\0&\lambda-4+a&\lambda-2\end{vmatrix}=\begin{vmatrix}\lambda-2&0&0\\1&\lambda-3&3\\0&\lambda-4+a&\lambda-2\end{vmatrix}$$

$$=(\lambda-2)\begin{vmatrix}\lambda&3\\2\lambda-6+a&\lambda-2\end{vmatrix}=(\lambda-2)[\lambda^2-2\lambda-6\lambda+18-3a]=(\lambda-2)(\lambda^2-8\lambda+18-3a)$$

① $=(\lambda-2)(\lambda-2)(\lambda-k)$

$$\lambda^2-\underline{(2+k)}\lambda+2k=\lambda^2\underline{-8}\lambda+18-3a$$

$$\begin{cases} 2+k=8 \rightarrow k=6 \\ 18-3a=2k \rightarrow 18-3a=12 \quad \underline{\underline{a=3}} \end{cases}$$

② $(\lambda-2)(\lambda^2-8\lambda+18-3a)=(\lambda-2)(\lambda-k)^2 \qquad (\lambda\neq 2)$

$$\lambda^2-2k\lambda+k^2=\lambda^2-8\lambda+18-3a$$

$$\Rightarrow \underline{\underline{a=\frac{2}{3}}}$$

抽象型行列式的计算

$$|A\cdot B|=|A||B|$$

例1.11

例 1.11 已知 4 阶行列式 $|\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\beta}|=a$，$|\boldsymbol{\beta}+\boldsymbol{\gamma},\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_1|=b$，则 $|\boldsymbol{\alpha}_2+\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_3,\boldsymbol{\gamma}|=$

$$\begin{aligned}
&|\alpha_2+\alpha_3,\alpha_1,\alpha_3,\gamma| \\
&=|\alpha_2,\alpha_1,\alpha_3,\gamma|+\underbrace{|\alpha_3,\alpha_1,\alpha_3,\gamma|}_{0} \\
&=|\alpha_2,\alpha_3,\gamma,\alpha_1| \\
&=|\gamma,\alpha_2,\alpha_3,\alpha_1| \\
&=|\beta+\gamma,\alpha_2,\alpha_3,\alpha_1|-|\beta,\alpha_2,\alpha_3,\alpha_1| \\
&=b+a
\end{aligned}$$

例1.12

例 1.12 设 $\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3$ 均为 3 维列向量，已知

$$\boldsymbol{A}=[\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3],\quad \boldsymbol{B}=[\boldsymbol{\alpha}_1-\boldsymbol{\alpha}_2+2\boldsymbol{\alpha}_3,2\boldsymbol{\alpha}_1+3\boldsymbol{\alpha}_2-5\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_1+2\boldsymbol{\alpha}_2-\boldsymbol{\alpha}_3],$$

且 $|\boldsymbol{A}|=2$，则 $|\boldsymbol{B}-\boldsymbol{A}|=$________.

$$|B-A|=|-a_2+2a_3,\ 2a_1+2a_2-5a_3,\ a_1+2a_2-2a_3|$$

$$\left(\begin{bmatrix}1&2&1\\-1&3&2\\2&-5&-1\end{bmatrix}=\begin{bmatrix}1&2&1\\0&5&3\\0&-9&-3\end{bmatrix}=\begin{bmatrix}1&2&1\\0&1&1\\0&0&6\end{bmatrix}=\begin{bmatrix}1&0&-1\\0&1&1\\0&0&6\end{bmatrix}=\begin{bmatrix}1&0&0\\0&1&0\\0&0&5\end{bmatrix}\right)$$

$$|B-A|=5|A|=10$$

$$A=[\alpha_1,\alpha_2,\alpha_3]\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$$

$$B=\boxed{[\alpha_1,\alpha_2,\alpha_3]\begin{bmatrix}1&2&1\\-1&3&2\\2&-5&-1\end{bmatrix}} \Rightarrow |B|=|\alpha_1\alpha_2\alpha_3|\left|\begin{matrix}1&2&1\\-1&3&2\\2&-5&-1\end{matrix}\right|=6|\alpha_1\alpha_2\alpha_3|$$

$$\Rightarrow |B-A|=10$$

余子式和代数余子式的线性组合的计算

由 $$a_{i1}A_{i1}+a_{i2}A_{i2}+\cdots+a_{in}A_{in}=\begin{vmatrix} & & * & \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ & & * & \end{vmatrix},\qquad ①$$

则 $$k_1A_{i1}+k_2A_{i2}+\cdots+k_nA_{in}=\begin{vmatrix} & & * & \\ k_1 & k_2 & \cdots & k_n \\ & & * & \end{vmatrix},\qquad ②$$ ← 只改一行

其中 * 处表示元素不变，①，②的区别仅仅在于第 i 行的元素 $a_{i1},a_{i2},\cdots,a_{in}$ 换成了 $k_1,k_2,\cdots,k_n$，这样，给出不同的系数 $k_1,k_2,\cdots,k_n$，即得到了不同的行列式. 而若要求 $k_1M_{i1}+k_2M_{i2}+\cdots+k_nM_{in}$，只需用 $M_{ij}=(-1)^{i+j}A_{ij}$ 化为关于 A_{ij} 的线性组合即可.

例1.13

例 1.13 设 $|\boldsymbol{A}|=\begin{vmatrix} 2 & -1 & 2 & 3 \\ 0 & 1 & -1 & 0 \\ 2 & 3 & 4 & 5 \\ 1 & 1 & 1 & 1 \end{vmatrix}$，则 $A_{31}+A_{32}+A_{33}+M_{34}=($ $)$.

$$\underset{1}{A_{31}}+\underset{1}{A_{32}}+\underset{1}{A_{33}}+\underset{-A_{34}}{M_{34}} = \begin{vmatrix} 2 & -1 & 2 & 3 \\ 0 & 1 & -1 & 0 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} 2 & -1 & 1 & 3 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 2 & -1 \\ 2 & 2 & 4 & 0 \end{vmatrix} = \begin{vmatrix} 2 & -3 & 3 \\ 1 & 0 & -1 \\ 2 & 0 & 0 \end{vmatrix} = 2\times 3 = 6$$

习1.1

$$D_4=\begin{vmatrix} 4 & 1 & 2 & 3 \\ 3 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 \\ 1 & 2 & 3 & 4 \end{vmatrix} = \begin{vmatrix} 10 & 1 & 2 & 3 \\ 10 & 4 & 1 & 2 \\ 10 & 3 & 4 & 1 \\ 10 & 2 & 3 & 4 \end{vmatrix} = 10\begin{vmatrix} 1 & 0 & 1 & 2 \\ 1 & 3 & 0 & 1 \\ 1 & 2 & 3 & 0 \\ 1 & 1 & 2 & 3 \end{vmatrix} = 10\begin{vmatrix} 1 & 0 & 1 & 2 \\ 0 & 3 & -1 & -1 \\ 0 & 2 & 2 & -2 \\ 0 & 1 & 1 & 1 \end{vmatrix}$$

$$= 10\begin{vmatrix} 4 & 0 & 0 \\ 2 & 2 & -2 \\ 1 & 1 & 1 \end{vmatrix} = 40\times 4 = 160$$

习1.2

n 阶行列式 $\begin{vmatrix} a & b & 0 & \cdots & 0 & 0 \\ 0 & a & b & \cdots & 0 & 0 \\ 0 & 0 & a & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & a & b \\ b & 0 & 0 & \cdots & 0 & a \end{vmatrix}_{n\times n} =$

$$a\cdot a^{n-1} + b\cdot(-1)^{n+1}\cdot b^{n-1} = a^n+(-1)^{n+1}b^n$$

习1.3

$$D_4=\begin{vmatrix} 1 & a & 0 & 0 \\ -1 & 2-a & b & 0 \\ 0 & -2 & 3-b & c \\ 0 & 0 & -3 & 4-c \end{vmatrix} = \begin{vmatrix} 1 & a & 0 & 0 \\ 0 & 2 & b & 0 \\ 0 & 0 & 3 & c \\ 0 & 0 & 0 & 4 \end{vmatrix} = 4! = 24$$

习1.4

$$D_n=\begin{vmatrix} 2 & -1 & 0 & \cdots & 0 & 0 \\ -1 & 2 & -1 & \cdots & 0 & 0 \\ 0 & -1 & 2 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 2 & -1 \\ 0 & 0 & 0 & \cdots & -1 & 2 \end{vmatrix}_{n\times n}.$$

$$=\begin{vmatrix} 1 & -1 & & & \\ 0 & 2 & -1 & & \\ 0 & -1 & 2 & \ddots & \\ \vdots & & \ddots & \ddots & -1 \\ 1 & & & -1 & 2 \end{vmatrix} = (-1)^{n-1}(-1)^{n+1} + D_{n-1}$$

$$= 1+D_{n-1} = n-1+D_1 = n+1$$

习1.5

计算 n 阶行列式 $D_n=\begin{vmatrix} a_1 & -1 & 0 & \cdots & 0 & 0 \\ a_2 & x & -1 & \cdots & 0 & 0 \\ a_3 & 0 & x & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a_{n-1} & 0 & 0 & \cdots & x & -1 \\ a_n & 0 & 0 & \cdots & 0 & x \end{vmatrix}.$

$$= x\cdot(-1)^{2n} D_{n-1} + a_n(-1)^{n-1}(-1)^{n+1}$$

$$= xD_{n-1} + a_n$$

$$D_n = a_n + xD_{n-1} = a_n + x(a_{n-1} + xD_{n-2})$$

$$= a_n + xa_{n-1} + x^2a_{n-2} + \cdots + x^{n-1}a_1$$

习1.6

已知 $\begin{vmatrix} \lambda & 2 & 2 \\ -2 & \lambda-2 & 2 \\ 2 & 2 & \lambda-2 \end{vmatrix}=0$，则 $\lambda=$__________.

$$\begin{vmatrix} \lambda & 2 & 2 \\ -2 & \lambda-2 & 2 \\ 0 & \lambda & \lambda \end{vmatrix} = \begin{vmatrix} \lambda & 2 & 0 \\ -2 & \lambda-2 & -\lambda+4 \\ 0 & \lambda & 0 \end{vmatrix} = -\lambda\cdot\lambda(-\lambda+4) = \lambda^2(\lambda-4)=0$$

$\lambda=0$ 或 $\lambda=4$

习1.7

一元二次方程 $\begin{vmatrix} x & 1 & a \\ 1 & x-1 & 1 \\ 1 & 1 & 1 \end{vmatrix}=0$ 有二重根，求参数 a 的值及该二重根.

$$\begin{vmatrix} x & 1 & a \\ 1 & x-1 & 1 \\ 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} x & 1-x & a-x \\ 1 & x-2 & 0 \\ 1 & 0 & 0 \end{vmatrix} = (-1)^{\frac{3\times2}{2}}(x-2)(a-x) = (x-2)(x-a)$$

$a=2$

习1.8

1.8 已知 $\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\beta},\boldsymbol{\gamma}$ 均为 4 维列向量，且

$$|\boldsymbol{\gamma},\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3|=n,\quad |\boldsymbol{\alpha}_1,\boldsymbol{\beta}+\boldsymbol{\gamma},\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3|=m,$$

则 $|\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,3\boldsymbol{\beta}|=$__________.

$$\left.\begin{aligned} |\alpha_1,\gamma,\alpha_2,\alpha_3| &= -n \\ |\alpha_1,\beta+\gamma,\alpha_2,\alpha_3| &= m \end{aligned}\right\} \quad \begin{aligned} |\alpha_1,\beta,\alpha_2,\alpha_3| &= m+n \\ |\alpha_1,\alpha_2,\alpha_3,3\beta| &= 3(m+n) \end{aligned}$$

习1.9

1.9 已知 $\alpha_1,\alpha_2,\alpha_3$ 是 3 维列向量，且 $|\alpha_1,\alpha_2,\alpha_3|\neq 0$，$A$ 是 3 阶矩阵，满足

$$A\alpha_1=\alpha_2-2\alpha_3,\quad A\alpha_2=\alpha_1-\alpha_2+2\alpha_3,\quad A\alpha_3=2\alpha_1+\alpha_2,$$

则 $|A|=$________.

$$\begin{bmatrix} A \end{bmatrix}\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}=\begin{bmatrix} 0 & 1 & 2 \\ 1 & -1 & 1 \\ -2 & 2 & 0 \end{bmatrix}\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{bmatrix}$$

$$|A|=\begin{vmatrix} 0 & 1 & 2 \\ 1 & -1 & 1 \\ -2 & 2 & 0 \end{vmatrix}=\begin{vmatrix} 0 & 1 & 2 \\ 1 & 0 & 1 \\ -2 & 0 & 0 \end{vmatrix}=-2$$

习1.10

1.10 设

$$|A|=\begin{vmatrix} -2 & 2 & -2 & 2 \\ 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 2 & -1 & 3 & 5 \end{vmatrix}.$$

计算：(1)$A_{41}+A_{42}+A_{43}+A_{44}$，其中 A_{4j} 是元素 $a_{4j}(j=1,2,3,4)$ 的代数余子式；(2)$M_{31}+M_{32}+M_{33}+M_{34}$，其中 M_{3j} 是元素 $a_{3j}(j=1,2,3,4)$ 的余子式.

(1) $A_{41}+A_{42}+A_{43}+A_{44}=\begin{vmatrix} -2 & 2 & -2 & 2 \\ 1 & 2 & 3 & 4 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{vmatrix}=0$

(2) $M_{31}+M_{32}+M_{33}+M_{34}$

$=A_{31}-A_{32}+A_{33}-A_{34}=\begin{vmatrix} -2 & 2 & -2 & 2 \\ 1 & 2 & 3 & 4 \\ 1 & -1 & 1 & -1 \\ 2 & -1 & 3 & 5 \end{vmatrix}=\begin{vmatrix} -2 & 0 & 0 & 0 \\ 1 & 3 & 7 & 5 \\ 1 & 0 & 0 & 0 \\ 2 & 1 & 8 & 7 \end{vmatrix}=0$

300.1.1

1.1 下列行列式中等于零的是（ ）.

(A) $\begin{vmatrix} 1 & 2 & 3 \\ -1 & 0 & 3 \\ 2 & 2 & 5 \end{vmatrix}$ (B) $\begin{vmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 2 & 2 & 3 \end{vmatrix}$

(C) $\begin{vmatrix} 1 & 2 & 3 \\ 0 & -4 & 0 \\ -2 & -7 & -6 \end{vmatrix}$ (D) $\begin{vmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -2 & 3 \end{vmatrix}$

(A) $\begin{vmatrix} 0 & 2 & 6 \\ -1 & 0 & 3 \\ 2 & 2 & 5 \end{vmatrix}=\begin{vmatrix} 0 & 2 & 6 \\ -1 & 0 & 0 \\ 2 & 2 & 11 \end{vmatrix}=22-12\neq 0$

(B) $\begin{vmatrix} 0 & 2 & 5 \\ -1 & 0 & 2 \\ 0 & 2 & 7 \end{vmatrix}\neq 0$

(C) $-4\begin{vmatrix} 1 & 3 \\ -2 & -6 \end{vmatrix}=0$ (C)

300.1.2

$$\begin{vmatrix} a & b & c & d \\ x & 0 & 0 & y \\ y & 0 & 0 & x \\ d & c & b & a \end{vmatrix} = \begin{vmatrix} c & b & a & d \\ b & c & d & a \\ 0 & 0 & y & x \\ 0 & 0 & x & y \end{vmatrix} = (c^2-b^2)(y^2-x^2)$$

300.1.3

$$\begin{vmatrix} -1 & -1 & -1 & -1 \\ -1 & -1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{vmatrix} = \begin{vmatrix} -2 & 0 & 0 & 0 \\ -1 & -1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ -1 & 1 & 1 & 1 \end{vmatrix} = -2\begin{vmatrix} 0 & 0 & 2 \\ -1 & 1 & 1 \\ 1 & 1 & 1 \end{vmatrix} -4(-1-1) = 8$$

300.1.4

$$\begin{vmatrix} 1 & b_1 & 0 & 0 \\ -1 & 1-b_1 & b_2 & 0 \\ 0 & -1 & 1-b_2 & b_3 \\ 0 & 0 & -1 & 1-b_3 \end{vmatrix} = \begin{vmatrix} 1 & b_1 & 0 & 0 \\ 0 & 1 & b_2 & 0 \\ 0 & 0 & 1 & b_3 \\ 0 & 0 & 0 & 1 \end{vmatrix} = 1$$

300.1.5

$\alpha_3\alpha_1\alpha_2\beta_2$

$\alpha_1\alpha_2\alpha_3\beta_2$

1.5 设 $\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\beta}_1,\boldsymbol{\beta}_2$ 都是 4 维列向量，且 4 阶行列式 $|\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\beta}_1|=m$，$|\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\beta}_2,\boldsymbol{\alpha}_3|=n$，则 4 阶行列式 $|\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_1,\boldsymbol{\beta}_1+\boldsymbol{\beta}_2|$ 等于(　　).

(A) $m+n$　　(B) $-(m+n)$　　(C) $n-m$　　(D) $m-n$

$|\alpha_3,\alpha_2\,\alpha_1,\beta_1| = -m$　　$|\alpha_3\ \alpha_2\ \alpha_1\ \beta_2| = n$

$\Rightarrow\ n-m\quad$ (C)

300.1.6

1.6 已知 4 阶行列式 $|\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\beta}|=a$，$|\boldsymbol{\beta}+\boldsymbol{\gamma},\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_1|=b$，其中 $\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_2,\boldsymbol{\alpha}_3,\boldsymbol{\beta},\boldsymbol{\gamma}$ 均为 4 维列向量，计算 $|\boldsymbol{\alpha}_1,\boldsymbol{\alpha}_3,\boldsymbol{\alpha}_2,\boldsymbol{\gamma}|$.

$|\alpha_1,\alpha_3,\alpha_2,\beta| = -a$

$|\alpha_1,\alpha_3,\alpha_2,\beta+\gamma| = b$

$\Rightarrow |\alpha_1,\alpha_3,\alpha_2,\gamma| = b+a$

300.1.7

$$\begin{vmatrix} -a_1 & 0 & 0 & 1 \\ a_1 & -a_2 & 0 & 1 \\ 0 & a_2 & -a_3 & 1 \\ 0 & 0 & a_3 & 1 \end{vmatrix} = \begin{vmatrix} -a_1 & 0 & 0 & 1 \\ 0 & -a_2 & 0 & 2 \\ 0 & 0 & -a_3 & 3 \\ 0 & 0 & 0 & 4 \end{vmatrix} = -4a_1a_2a_3$$

300.1.8

已知方程 $\begin{vmatrix} x-2 & 0 & 0 \\ -3 & x-1 & a \\ 2 & a & x-1 \end{vmatrix}=0$ 有二重根，求满足条件的常数 a 及方程的根.

$$(x-2)\begin{vmatrix} x-1 & a \\ a & x-1 \end{vmatrix} = (x-2)(x^2-2x+(1-a^2))$$

① $(x-2)(x-k) = x^2-(2+k)x+2k$　　$k=0$

$= x^2 - 2x + (1-a^2)$　　$a^2=1 \Rightarrow a=\pm 1$

② $(x-k)^2 = x^2-2kx+k^2$　　$k=1$

$= x^2-2x+(1-a^2)$　　$a=0$

300.1.9

1.9 设 $A=[\alpha_1,\alpha_2,\alpha_3]$是 3 阶矩阵，且$|A|=4$，若

$$B=[\alpha_1-3\alpha_2+2\alpha_3,\alpha_2-2\alpha_3,2\alpha_2+\alpha_3],$$

则$|B|=$________.

$$|B|=\left|[\alpha_1,\alpha_2,\alpha_3]\begin{bmatrix}1&0&0\\-3&1&2\\2&-2&1\end{bmatrix}\right|=4\begin{vmatrix}1&2\\-2&1\end{vmatrix}=20$$

300.1.10

1.10 设 $D=\begin{vmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\a_{31}&a_{32}&a_{33}\end{vmatrix}$，$A_{ij}$为 D 的(i,j)元的代数余子式，则 $A_{31}+2A_{32}+3A_{33}=$(A).

(A) $\begin{vmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\1&2&3\end{vmatrix}$　(B) $\begin{vmatrix}a_{11}&a_{12}&a_{13}\\a_{21}&a_{22}&a_{23}\\1&-2&3\end{vmatrix}$

(C) $\begin{vmatrix}a_{11}&a_{12}&1\\a_{21}&a_{22}&2\\a_{31}&a_{32}&3\end{vmatrix}$　(D) $\begin{vmatrix}a_{11}&a_{12}&1\\a_{21}&a_{22}&-2\\a_{31}&a_{32}&3\end{vmatrix}$